CS3700 — Introduction to Database Systems

Tier	A
Phase (order)	4
Do this first	CS2705, CS1200
Estimated effort	10-12 weeks
Unlocks	CS6550, CS6510, CS6220, DA3406

Roadmap

1. ER modelling and the relational model; relational algebra.
2. SQL: select-from-where, joins, nested queries, aggregates.
3. Functional dependencies and normal forms (1NF-BCNF), decomposition.
4. Transactions, indexing, concurrency control, recovery.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Introduction to Database Systems (41 videos). NPTEL-NOC IITM *Introduction to Database Systems* (IIT Madras).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Database Management Systems (81 videos). IIT Madras BS *Database Management Systems* (Partha Pratim Das).
- CMU Intro to Database Systems (15-445/645 - Fall 2024) (26 videos). CMU 15-445 Fall 2024 (Andy Pavlo, official CMU Database Group): how a DBMS is built inside. Do BusTub projects.

⚡ Short videos for intuition and revision

- SQL Ultimate Course (56 videos). Data with Baraa *SQL Ultimate Course*: practical SQL from zero.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–3	DB architecture. Install PostgreSQL; load a sample DB (e.g. pagila).
2	4–7, 12	ER modelling, weak entities, relational model, keys, ER -> tables. Design an ER schema for a college and create it in Postgres.

Week	Lectures	What to do
3	8–14	Relational algebra and tuple relational calculus. Write 15 queries in RA, then in SQL.
4	15–20	SQL: DDL, subqueries, correlated subqueries, aggregates, views, programmatic access. Do 30 SQL problems (pgexercises.com / LeetCode SQL 50).
5	21–26	Functional dependencies, Armstrong's axioms, 2NF/3NF/BCNF/4NF, decompositions. Normalise 5 schemas by hand.
6	27–35	File organisation, hashing, B+ trees, RAID, query evaluation, join algorithms, optimisation. Use EXPLAIN ANALYZE on your queries; add indexes and compare.
7	36–41	Transactions, schedules, locking, undo/redo logging, isolation levels. Reproduce a lost-update anomaly in two psql sessions.

▼ **All lectures in *Introduction to Database Systems* (41 videos)**

1. #1 Introduction to Database Systems
2. #2 Database Architecture | Introduction to Database Systems
3. #3 RDBMS Architecture | Introduction to Database Systems
4. #4 Introduction to ER Model | Introduction to Database Systems
5. #5 Entities & Relationships | Introduction to Database Systems
6. #6 Modelling Weak Entities & Design Choices | Introduction to Database Systems
7. #7 Relational Data Model & Notion of Keys | Introduction to Database Systems
8. #8 Introduction to Relational Algebra | Introduction to Database Systems
9. #9 Operators in Relational Model | Introduction to Database Systems
10. #10 Uses of Renaming, Join & Division in Relation Algebra | Introduction to Database Systems
11. #11 Example Queries in Relational Model & Outer Join Operation | Introduction to Database Systems
12. #12 Convert ER-Model to a Relational Model | Introduction to Database Systems
13. #13 Introduction to Tuple Relational Calculus | Introduction to Database Systems
14. #14 Example TRC Queries | Introduction to Database Systems
15. #15 Data Definition using SQL | Introduction to Database Systems
16. #16 Basic SQL Query block & Subqueries | Introduction to Database Systems
17. #17 Correlated Subqueries | Introduction to Database Systems
18. #18 Aggregate Functions | Introduction to Database Systems
19. #19 Views | Introduction to Database Systems
20. #20 Programmatic Access of SQL | Introduction to Database Systems
21. #21 Normal forms | Introduction | Introduction to Database Systems
22. #22 Deriving New Functional Dependencies | Introduction to Database Systems
23. #23 Proving Soundness & Completeness of Armstrong's Axioms | Introduction to Database Systems
24. #24 Normal forms | 2NF, 3NF, BCNF | Introduction to Database Systems
25. #25 Properties of Decompositions | Introduction to Database Systems
26. #26 Normal forms | 4NF, 5NF | Introduction to Database Systems
27. #27 Introduction to File Orgranization | Introduction to Database Systems

28. #28 File Orgranization Methods | Introduction to Database Systems
29. #29 Dynamic File Orgranization Using Hashing | Introduction to Database Systems
30. #30 Index Structures | Introduction to Database Systems
31. #31 B+Trees on Disks | Introduction to Database Systems
32. #32 Performance & Reliability of Multiple Disks | Introduction to Database Systems
33. #33 Relational Query Evaluation | Introduction to Database Systems
34. #34 Join Operator Processing Algorithms | Introduction to Database Systems
35. #35 Query Optimization | Introduction to Database Systems
36. #36 ACID Properties & Operations in Transactions | Introduction to Database Systems
37. #37 Schedules | Introduction to Database Systems
38. #38 Concurrency Control Using Locks | Introduction to Database Systems
39. #39 Recovery Using Undo Logging Method | Introduction to Database Systems
40. #40 Recovery Using Redo & Undo-Redo Logging Methods | Introduction to Database Systems
41. #41 Recoverable Schedules & Transaction Isolation Levels | Introduction to Database Systems

Books, notes and websites

- ✅ NPTEL *Database Management System* (Partha Pratim Das, IIT Kharagpur) — onlinecourses.nptel.ac.in/noc21_cs58/preview
- CMU 15-445 *Database Systems* (Andy Pavlo) — YouTube
- Book: Silberschatz, Korth, Sudarshan, *Database System Concepts*
- Practice: SQLBolt, pgexercises.com, PostgreSQL docs

Practice / proof of understanding

Design a normalised schema for a small app, load sample data in PostgreSQL, write 25 queries, and prove one decomposition lossless.

What your IITM course sheet says

From the PDF: CS3700

Official description: The objective of the course is to impart knowledge about fundamental ideas and techniques of relational database systems.

Official syllabus (trimmed):

> Introduction - General introduction to database systems; Database - DBMS disctinction, approaches to building a database, data models, database management system, three-schema architecture of a database, challenges in building a DBMS, various components of a DBMS.E/R Model - Conceptual data modeling - motivation, entities, entity types, various types of attributes, relationships, relationship types, E/R diagram notation, examples.Relational Data Model - Concept of relations, schema-instance distinction, keys, referential integrity and foreign keys, relational algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relation calculus, converting the database specification in E/R notation to the relational schema.SQL - Introduction, data definition in SQL, table, key and foreign key definitions, update behaviors. Querying in SQL - basic select-from-where block and its semantics, nested queries - correlated and uncorrelated, notion of aggregation, aggregation functions group by and having clauses, embedded SQL.Dependencies and Normal forms - Importance of a good schema design, problems encountered with bad schema designs, motivation for normal forms, dependency theory - functional dependencies, Armstrong's axioms for FD's, closure of a set of FD's, minimal covers, definitions of 1NF, 2NF, 3NF and BCNF, decomposit

Official textbooks: 1. Ramez Elmasri and Shamkant B Navathe, Fundamentals of Database Systems, 6th Edition, Pearson, 2013

Official references: 1. A Silberschatz, H F Korth and S Sudarshan, Database System Concepts, 6th Edition, McGraw Hill, 2011. 2. H Garcia-Molina, J D Ullman, and Jennifer Widom, Database Systems-The Complete Book, Pearson Education, 2004. 3. Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, 3rd Edition, McGraw Hill, 2003.